

Project Proposal: GPU-Accelerated Particle Simulation

1. Application/Problem Description:

Particle simulations are widely used in computer science applications such as real-time graphics, physics engines, scientific modeling, and animation. These systems simulate the motion of a large number of particles governed by the rules of physics.

A key challenge in particle simulation is scalability; on a CPU, updating each particle sequentially becomes a bottleneck as the number of particles increases. Real-time performance is difficult to maintain when simulating hundreds, to thousands, to millions of particles.

This project aims to address this limitation by leveraging GPU computing to accelerate particle updates. Since each particle can be updated independently at each timestep, the problem is highly parallelizable and well-suited for execution on a GPU.

2. Proposed Approach/Implementation Plan:

The project will implement both CPU and GPU versions of a particle simulation and compare their performance.

The idea is that each particle stores position and velocity, and at each timestep the system applies gravity, updating the particles' velocity and position, and handles boundary collisions.

A CPU baseline will first be developed in C++ using a sequential loop to update all particles. This will be utilized to establish correctness and provide a performance reference.

The GPU implementation will use CUDA to parallelize computation. Particle data will be stored in a structure of arrays format for efficient memory access. A CUDA kernel will assign one thread per particle to perform physics updates in parallel.

For visualization, particles will be rendered in real time using OpenGL with GLFW, displaying them as points.

3. Planned Demo:

The final demo will demonstrate the benefits of GPU acceleration with the following components:

- a. Real-Time Simulation
 - i. Display particles moving under gravity in a bounded environment
- b. CPU vs GPU Comparison

- i. Run identical simulations on CPU and GPU to compare frames per second and execution time per frame
- c. Scalability Analysis
 - i. Show how performance scales for CPU vs GPU as particle count increases
- d. Performance Metrics
 - i. Report on the overall performance of the particle simulation in terms of speedup factor between the CPU and GPU, frame time consistency, and maximum real-time particle count